

KARAN

meena.karan9k@gmail.com • +44 7543 814 202 • UK (Full Work Authorisation)
linkedin.com/in/karan-th • github.com/Karanm5

PROFILE

Data professional with an MSc in Data Analytics (Distinction) and commercial experience building data systems that bridge the gap between raw information and actionable insight. My work spans data engineering (ETL pipelines, cloud architecture, data quality frameworks), machine learning (ensemble models, reinforcement learning, explainability), and analytics (dashboards, stakeholder reporting, data storytelling). I approach problems with academic rigour and commercial pragmatism: caring equally about technical correctness and whether the systems I build are genuinely useful to the people who rely on them. Currently preparing research for arXiv publication.

TECHNICAL SKILLS

Programming and Analysis: Python (pandas, NumPy, scikit-learn), SQL (advanced), R (fundamentals), applied statistics, data modelling

Machine Learning: scikit-learn, TensorFlow/Keras, PyTorch, XGBoost, LightGBM, ensemble methods, reinforcement learning (PPO), SHAP explainability, model evaluation and validation

Data Architecture: Snowflake, Delta Lake, medallion architecture (Bronze/Silver/Gold), ETL/ELT pipeline development, database design, data quality frameworks

Cloud and Infrastructure: AWS, Google Cloud Platform, BigQuery, Apache Airflow (certified), Apache Kafka (certified), Docker, CI/CD, FastAPI

Visualisation and Reporting: Tableau (proficient), Power BI, dashboard development, data storytelling, stakeholder communication

EDUCATION

MSc Data Analytics (Distinction) | Aston University, Birmingham | 2025

Modules: Statistical Machine Learning, Artificial Neural Networks, Probabilistic Modelling, Data Engineering, Data Preparation

Dissertation: Investigated the impact of disorder on superconductors' critical temperatures using machine learning. Applied Random Forest, Gaussian Process Regression, Monte Carlo simulation, and Variational Autoencoders to synthetic BCS data. Awarded Distinction.

Research currently being prepared for arXiv publication. Supervisor: Prof. Igor Yurkevich

BE Information Technology | Netaji Subhas Institute of Technology (NSIT), University of Delhi | 2022

PROFESSIONAL EXPERIENCE

Data Analyst | GigaTasking | February 2022 – April 2023

Designed and maintained ETL pipelines processing 50,000+ daily transactional records, ensuring data accuracy through systematic validation and quality checks

Developed automated workflows that reduced manual data handling by 40%, requiring stakeholder engagement across Operations, Finance, and IT to drive adoption

Created dashboards and analytical reports using Tableau, translating complex data into clear insight for senior leadership and cross-functional teams

Collaborated with business stakeholders to gather requirements and translate business problems into technical specifications

Mentored junior colleagues on SQL and Python, contributing to team capability development

SELECTED PROJECTS

Credit Risk Intelligence Platform | Snowflake, Delta Lake, XGBoost, TensorFlow, FastAPI, SHAP

Designed and built a production-grade data platform implementing medallion architecture (Bronze/Silver/Gold layers) for real-time credit risk assessment. Integrated ensemble ML models (XGBoost, LightGBM, Neural Networks) with

SHAP-based explainability. Deployed via FastAPI with CI/CD automation, demonstrating end-to-end delivery from data ingestion to production serving.

Reinforcement Learning Trading Agent / *PyTorch, PPO, Gymnasium, Custom Reward Engineering*

Developed a PPO-based RL agent in PyTorch, building custom policy and value networks from scratch. Engineered a novel risk-adjusted reward function penalising drawdowns, volatility, and overtrading. Includes custom Gymnasium environment with realistic portfolio constraints and comprehensive backtesting pipelines.

Real-Time Financial Data Pipeline / *Apache Kafka, Python, Streaming, MLOps*

Designed end-to-end streaming pipeline for real-time financial data ingestion and processing using event-driven architecture with Apache Kafka. Implemented automated evaluation pipelines and reproducible metrics, demonstrating MLOps best practices.

Biomedical Data Analytics Pipeline / *Python, SQL, Data Validation, Documentation*

Developed analytical pipeline for complex biomedical dataset (49,000 rows, 1,250 features) with rigorous attention to data quality and validation. Implemented systematic checks (type, range, cross-field consistency) and created comprehensive documentation including data dictionaries and schema definitions.

CERTIFICATIONS

Snowflake for Developers and Data Engineers (2026) • ETL and Data Pipelines with Airflow and Kafka, IBM (2026) • AWS Cloud Technical Essentials (2025) • DevOps, DataOps and MLOps Fundamentals (2025) • Introduction to Large Language Models, Google (2025)

ADDITIONAL INFORMATION

Availability: Available immediately. Full UK work authorisation (Graduate Visa).

Research: Currently preparing MSc dissertation on ML for superconductivity for arXiv publication.

References: Prof. Igor Yurkevich (Aston University), Dr. Philip Trevelyan (Aston University), Dr. Amarjit (NSIT). Letters available upon request.

Portfolio: github.com/Karanm5